Supplemental Material for "Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch"

Supplemental Material

Citation numbers follow the numbering of the main article.

1 Native-Sample Safety and QP Diagnostics

All benchmark violation rates are evaluated at the native 1 s controller instants. A controller sample is counted once when any enforced pressure, separator-enthalpy, or power barrier satisfies $\min_j h_j(x_k) < -10^{-2}$ in raw barrier units. The tolerance excludes numerical boundary roundoff; the unrounded minimum barrier is retained separately. Episode-level event probabilities are not combined with these sample rates.

Table S1. Count-level S3 diagnostics for the primary $\Delta g = 0$ validation. Each row aggregates five 500-sample rollouts. A rejected QP is non-converged, non-finite, or has maximum normalized primal residual above 10^{-6}.

Method	Violating samples	QP rejected	QP intervention
HOCBF, no GP	1682 of 2500	0 of 2500	4.48%
GP-HOCBF, $\epsilon_\kappa = 0$	36 of 2500	0 of 2500	67.08%
RoCBF-SF, $\epsilon_\kappa = 0.02$	0 of 2500	4 of 2500	67.12%
Full implemented-margin endpoint, $\epsilon_\kappa = 1$	419 of 2500	628 of 2500	51.12%

The four rejected $\epsilon_\kappa = 0.02$ candidates occur at the initial step of four seeds; the live upstream proposal is used on those cycles, and no subsequent state-limit violation is observed. At $\epsilon_\kappa = 1$, 628 rejected QPs coincide with the actuator-limited loss of full-row feasibility. This endpoint does not satisfy the robust-QP-feasibility assumption of Theorem 1.

2 Robustness-Factor Tune/Test Separation

Tune seeds 0–2 each contain 10 episodes of 500 controller samples. Thus, each ϵ_κ setting contains 15,000 tune samples. The predeclared selector is the smallest setting for which both pooled and maximum-per-seed violation rates are below 1% and both pooled and maximum-per-seed QP-rejection rates are below 0.5%.

Table S2. Tune-seed selection data. Rates are percentages of native controller samples. Wilson upper limits are descriptive Bernoulli-reference values; adjacent samples are serially correlated.

ϵ_κ	Violation count (rate)	Max. seed violation	QP rejection count (rate)	Max. seed rejection	Pass
0	412 (2.747)	8.240	0 (0.000)	0.000	No

ϵ_κ	Violation count (rate)	Max. seed violation	QP rejection count (rate)	Max. seed rejection	Pass
0.02	19 (0.127)	0.380	20 (0.133)	0.160	Yes
0.05	0 (0.000)	0.000	26 (0.173)	0.200	Yes
0.10	0 (0.000)	0.000	27 (0.180)	0.200	Yes
0.20	0 (0.000)	0.000	92 (0.613)	1.460	No
0.50	204 (1.360)	3.580	439 (2.927)	5.900	No
1.00	1047 (6.980)	11.080	1792 (11.947)	17.220	No

The selector fixes $\epsilon_\kappa = 0.02$ before evaluating seeds 3–4. Each held-out seed contains 25,000 samples. Seed 3 records 42 violations and 34 rejected QPs; seed 4 records 24 violations and 38 rejected QPs. The pooled rates are therefore 66/50,000 (0.132%) and 72/50,000 (0.144%), respectively. No holdout result is used to change ϵ_κ.

3 GP Residual Definition and Commissioning Data

The benchmark and deployed GP use the same three-dimensional kernel input,

$$x_k^{\mathrm{GP}} = [p_{m,k}, h_{m,k}, N_{e,k}]^\top. \tag{S1}$$

The nominal predictor uses the complete five-state vector and all three proposed control commands. The three learning targets are residual rates,

$$y_{j,k+1}^{\mathrm{GP}} = \frac{x_{j,k+1} - \hat{x}_{j,k+1|k}^{\mathrm{nom}}}{T_s}, \qquad j \in \{p_m, h_m, N_e\}. \tag{S2}$$

Kernel distances use the frozen training transform $(x_i - \mu_i^{\mathrm{train}})/s_i^{\mathrm{train}}$, and each residual target is standardized by its training mean and standard deviation. The transforms are not updated online.

The plant commissioning record contains approximately 14 days of 1 Hz historian and commissioning data (about 1.21 million state transitions). Stops, manual operation, communication faults, stale measurements, interlock action, missing values, and identified bad points are removed, leaving approximately 250,000 candidate transitions. Days 1–10 form the training candidate pool; day 11 is a 24 h isolation gap; and days 12–14 form the independent validation pool. Normalization parameters and kernel hyperparameters use only days 1–10.

Table S3. Fixed plant-GP load strata and quotas. Candidates are separated by at least 60 s and selected by deterministic farthest-point coverage in standardized (p_m, h_m, N_e) space. Quotas are not borrowed between strata.

Generator-load stratum (MW)	Training points	Validation points
180–264	100	40
264–363	100	40
363–462	100	40
462–561	100	40

Generator-load stratum (MW)	Training points	Validation points
561–660	100	40
Total	500	200

Three independent exact Matérn-5/2 GPs are trained and frozen after offline replay. The deployed dictionary remains at $N = 500$; online sample growth is zero. Models are reviewed quarterly and after major maintenance or a material coal-source change. The current and previous stable versions are retained for rollback. Exact-GP storage is $\mathcal{O}(qN^2)$ and offline retraining is $\mathcal{O}(qN^3)$ for $q = 3$ outputs. Three float32 Cholesky factors occupy approximately 3 MB; training arrays, coefficients, and kernel parameters occupy approximately 10–20 MB. The deployed JAX process has a larger runtime resident set (approximately 0.5–2 GB), but this does not grow with online operation because the dictionary is frozen.

The controlled sensitivity experiment uses five independent seeds. Each of the nine dictionary-size–corruption settings therefore comprises five seeded fit-and-evaluation runs (45 runs in total). Each run fits three scalar-output GPs, one for each component of $[p_m, h_m, N_e]$, giving 135 scalar GP fits. It then evaluates 200 held-out transitions and one 500-sample S3 closed-loop rollout. Corruption replaces 5% or 10% of selected training targets by signed three-training-standard-deviation offsets; it does not represent a measured plant bad-point rate.

Table S4. GP quantity/quality sensitivity. NRMSE and interval coverage are averaged over outputs and seeds. Closed-loop columns aggregate 2500 native controller samples per cell.

N	Target corruption	Held-out NRMSE	95% interval coverage	Violation, $\epsilon_\kappa = 0$	Violation, $\epsilon_\kappa = 0.02$	QP rejection, $\epsilon_\kappa = 0.02$
100	0%	0.00184	100.00%	7.60%	0.00%	0.16%
250	0%	0.00208	100.00%	7.92%	0.00%	0.16%
500	0%	0.00243	100.00%	6.96%	0.00%	0.16%
100	5%	0.43547	96.87%	1.08%	64.96%	95.64%
250	5%	0.37980	95.37%	1.32%	66.28%	98.64%
500	5%	0.25700	97.33%	1.12%	66.88%	99.44%
100	10%	0.49337	97.63%	1.28%	66.48%	98.76%
250	10%	0.49015	95.10%	1.04%	67.00%	99.36%
500	10%	0.42205	96.80%	0.68%	67.20%	99.88%

The corrupted fits are deliberately evaluated without the pre-enable data-quality gate to expose the failure mode. Their inflated uncertainty tightens the QP until most candidates are rejected; the controller then uses the upstream proposal, producing the 64.96–67.20% violation rates in Table S4. In deployment, the held-out residual and QP-feasibility checks reject these models before they receive write authority.

4 Coverage Monitoring and Fallback State Machine

The online input-coverage statistic and standardized innovation are

$$z_{i,k} = \frac{x_{i,k}^{\mathrm{GP}} - \mu_{x_i}^{\mathrm{train}}}{s_{x_i}^{\mathrm{train}}}, \tag{S3}$$

$$e_{j,k+1} = \frac{|r_{j,k+1} - \mu_j(x_k^{\mathrm{GP}})|}{\sqrt{\sigma_j^2(x_k^{\mathrm{GP}}) + \sigma_{n,j}^2}}, \quad r_{j,k+1} = \frac{x_{j,k+1} - \hat{x}_{j,k+1|k}^{\mathrm{nom}}}{T_s}.$$

Here $r_{j,k+1}$, μ_j, σ_j, and $\sigma_{n,j}$ all have residual-rate units. Because the deployed sampling period is $T_s = 1$ s, this notation correction does not change the reported numerical thresholds.

Table S5. GP coverage and degradation thresholds used by the deployed supervisor.

Monitor	Trigger	Action
Input coverage	Any $\lvert z_i \rvert > 3$ or outside the training 0.5th–99.5th percentile	Enter full-row nominal-HOCBF mode.
Posterior uncertainty	Above the validation-set 99th percentile	Withdraw GP mean correction.
Standardized innovation	Any $e_j > 3$ for three cycles	Withdraw GP mean correction and alarm.
Severe innovation	Any $e_j > 5$ once	Immediate live-DCS bypass.
Persistent OOD	More than 1% of cycles in one hour	Create an offline recalibration task.
Recovery	Ten consecutive cycles with valid input, innovation, and variance checks	Permit GP-mode recovery subject to controller authorization.

Moderate coverage loss follows

$$\begin{aligned} &\text{GP-RoCBF-SF} \rightarrow \text{nominal HOCBF} \quad (\text{for at most 10 cycles}),\\ &\text{nominal HOCBF} \rightarrow \text{GP-RoCBF-SF} \quad (\text{after 10 healthy cycles}),\\ &\text{nominal HOCBF} \rightarrow \text{BYPASS} \rightarrow \text{live DCS} \quad (\text{otherwise}). \end{aligned} \tag{S4}$$

Invalid measurements or communication, non-finite/model-load faults, nominal-QP or actuator-constraint infeasibility, a direct physical-bound exceedance, pressure-low physical margin below 2.0 MPa, or a QP/task timeout bypasses the bounded degraded stage immediately. The nominal-HOCBF interval has no GP coverage claim.

An initially infeasible QP permits one reduced-QP retry only for the whitelisted pressure_low row. Let $v = S_u\tilde{v}$, $\widetilde{G}_i = G_iS_u$, and $a_i = \| \widetilde{G}_i \|_2$. Row removal requires

$$i = \text{pressure\_low}, \quad a_i < 0.01, \quad h_i < -10^{-6}, \quad PM - PLO \geq 2.0\ \text{MPa}, \tag{S5}$$

with valid measurements and no active protection or interlock. The authority norm is evaluated after actuator normalization but before unit-row scaling. No actuator box/rate row or second CBF row can be removed, and only one retry is permitted. A successful reduced QP is reported as optimal_recovered; it is outside the full-row certificate.

The deployed task deadline is 1000 ms, equal to the controller period. The field QP timeout is 200 ms. A timeout, non-finite result, or failed retry returns the current live DCS command rather than the previous algorithm command. Three consecutive solve failures, communication loss beyond 3 s, a process restart, or a model-load fault latches bypass. Re-entry requires valid measurements, heartbeats, and dry-run QPs for ten cycles, DCS automatic mode, inactive protection/interlocks, operator reset acknowledgement, and the active Kubernetes lease. A 10 s linear blend is then used for bumpless transfer, provided the QP and DCS commands differ by no more than 2% of actuator range and both lie in the current feasible set.

Kubernetes uses a 3 s Lease renewed every 1 s. Only the lease holder may assert the local write permit; standby Pods connect to Modbus TCP read-only. The DCS remains the final write-permission owner and requires enable, automatic mode, no manual mode, valid echoed heartbeat with age no greater than 2 s, active lease ownership, and a fresh command sequence. Loss of any condition selects the live coordinated-control command.

5 Plant Historian and Controller-Export Evidence

The public production-evidence index separates historian context from controller execution. MW01–MW03 are post-retrofit low-to-mid-load historian windows used only to position measured load and main-steam pressure in the operating-envelope figure; they do not supply controller-internal QP fields. MW04–MW06 are confirmed two-hour plant-controller exports covering 480.7–629.7 MW. Each contains 1440 records at 5 s export spacing from a controller executing at 1 s, giving 4320 exported records rather than 4320 controller cycles. These exports directly support the timestamps, operating mode, QP/recovery status, logged margins, saturation flags, and timing statistics reported below. GP lifecycle, Kubernetes lease ownership, heartbeat/sequence checks, and DCS write-permission rules are deployment-configuration specifications; the exports do not expose every internal handshake transition.

The confirmed controller/historian mappings used in the revision are NE ↔ 20CQTP_MW, PM ↔ DCS2_20HAG10CP101, and HM ↔ DCS2_SEPARATOUT_ENTH. The independent tag DCS2_MAIN_PRESS is main-steam pressure. The exported PST_HI value is a main-steam-pressure replay bound, not the separator-pressure upper limit.

Table S6. High-load controller-export evidence. Direct margins are minima over all 1440 records in each window. "None" denotes no fallback or saturation event.

Window	Load (MW)	Sep.-pressure low margin (MPa)	Main-pressure high margin (MPa)	Nearest enthalpy margin (kJ/kg)	QP/recovery record	QP/task p95 (ms)
MW04	563.0–621.2	2.217	2.173	62.284	1397 optimal; 43 recovered; 1 fuel-sat.	4.195 / 26.238
MW05	547.9–629.7	3.438	1.483	20.354	1440 optimal; none	4.074 / 26.139
MW06	480.7–619.3	2.705	0.463	0.359	1440 optimal; none	4.062 / 26.159

All 43 MW04 recovery records have FB_RSN=weak_authority_constraint_drop, ACT_CSTR=pressure_low, and negative MG_PLO. The diagnostic row ranges from -0.03266 to -0.00561, while the direct physical margin $PM - PLO$ ranges from 2.21654 to 2.36306 MPa and the other five exported HOCBF margins remain positive. MW05–MW06 contain no reduced-QP record.

Across MW04–MW06, QP time has median/p95/maximum values 3.557/4.095/6.234 ms, and total task time has corresponding values 24.564/26.183/28.665 ms.

The plant value $\epsilon_\kappa = 0.1$ is fixed after training-block replay and passes the one-pass time-isolated validation gate without retuning; it is not copied from the benchmark S3 selection. Record-level plant replay remains a restricted enterprise asset and is not inferred from the high-load controller exports. The original PID gains, output limits, anti-windup, sliding-pressure curve, coordinated-control logic, and fuel/feedwater/turbine-valve base loops are unchanged; RoCBF-SF is added downstream with enable, bypass, heartbeat, command-quality, and DCS-selection logic. The outage also includes instrument calibration, actuator inspection, combustion-equipment maintenance, and heat-surface cleaning, so the historian before/after comparison is observational. The controller exports separately establish that the proposed filter executed after restart.

The public reproducibility package contains source code, simulation results, derived historian metrics, bounded controller-export excerpts, field maps, and source/public hashes. Complete raw historian and original controller exports are enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and the applicable data-use conditions.

6 Computation and Sampled-Data Boundary

The NMPC reference uses a warm-started SLSQP solve with horizon $N = 5$, output weights diag($1,0.001,0.01$), input weights diag($0.01,0.01,0.01$), deviation-input bounds $[-10,10]^3$, six prediction constraints, additive disturbance correction with gain 0.5, maxiter=50, and ftol= 10^{-4}. All 17,500 solves converge and pass the 10^{-6} prediction-residual check. Mean per-step solve time ranges from 3.50 ms in the nominal case to 31.99 ms in the tested perturbation scenarios. These values describe the implemented reference rather than an optimized industrial MPC package.

The theorem is continuous-time. The controller applies zero-order hold with $T_s = 1$ s because 1 s is the deployed coordinated-control task period; the safety filter therefore evaluates each upstream command update before the command-selection stage. This implementation choice does not assign a 1 s time constant to the slower boiler thermal dynamics. The field exports retain one record every 5 s, so their spacing is a logging interval rather than the internal control period. Field task timing establishes computational fit within the 1 s period but does not establish inter-sample forward invariance. Theorem 1 therefore is not presented as a sampled-data certificate; deriving such a certificate is retained as future work.

7 Proofs of Theoretical Results

7.1 Residual Cross-Term Bound

Lemma S1 (Residual cross-term bound). Let $\delta_i = \psi_i - \widehat{\psi}_i^0$ with $\delta_0 \equiv 0$. On the simultaneous residual event in Assumption 2 of the main article, suppose a valid pointwise derivative bound $\| \nabla_x \delta_{i-1}(x) \|_2 \leq G_\delta^{(i-1)}(x)$ is available on $\mathcal{X}_{\mathrm{op}}$. Then

$$|L_{\Delta \hat{f}} \delta_{i-1}(x)| \leq G_\delta^{(i-1)}(x) \beta \, \| \sigma_{\mathrm{GP}}(x) \|_2. \tag{S6}$$

Proof. By Cauchy–Schwarz,

$$|L_{\Delta\hat{f}}\delta_{i-1}| = |\nabla_x\delta_{i-1}^{\mathsf{T}}\Delta\hat{f}| \leq \| \nabla_x\delta_{i-1}\|_2 \, \| \Delta\hat{f}\|_2. \tag{S7}$$

The simultaneous residual event gives $\| \Delta\hat{f}(x)\|_2 \leq \beta \, \| \sigma_{\mathrm{GP}}(x)\|_2$ over the modeled outputs, which yields Eq. (S6). Smoothness on a compact operating set ensures that finite global derivative bounds exist when the required derivatives are bounded, but it does not identify them with $\| \nabla\hat{\psi}_{i-1}^{0}\|_2$. □

The online code uses $\| \nabla\hat{\psi}_{i-1}^{0}\|_2$ as a commissioning proxy for $G_\delta^{(i-1)}$. Theorem 1 applies to that implementation only when Assumption 3 of the main article independently verifies that the resulting total margin dominates the true top-level HOCBF residual on $\mathcal{X}_{\mathrm{op}}$.

7.2 Full Proof of Theorem 1

Proof of Theorem 1. **Step 1 (Simultaneous residual event).** Assumption 2 of the main article states that $|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \leq \beta\sigma_{\mathrm{GP},j}(x)$ holds simultaneously for the $q = 3$ modeled residual outputs and all $x \in \mathcal{X}_{\mathrm{op}}$ with probability at least $1 - \delta$. Condition on this event. The commissioning multiplier used in the numerical studies does not establish this event by itself.

Step 2 (Compositional bound). We show $|\delta_i(x)| \leq \sigma_i(x)$ by induction on i.

Base case ($i = 1$): $\delta_1 = \sum_j (\partial h/\partial x_j)\Delta\hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \leq \beta\sum_j |\partial h/\partial x_j|\sigma_{\mathrm{GP},j} = \sigma_1$.
Inductive step ($i \geq 2$): Assume $|\delta_{i-1}| \leq \sigma_{i-1}$. The residual recursion in the main article gives $|\delta_i| \leq |L_{\Delta\hat{f}}\hat{\psi}_{i-1}^{0}| + (|L_{\hat{f}}|_{\mathrm{op}} + k_i)|\delta_{i-1}| + |L_{\Delta\hat{f}}\delta_{i-1}|$. The first term is bounded as in the base case. The second term is bounded by the induction hypothesis. Lemma S1 bounds the third term when a valid $G_\delta^{(i-1)}$ is available. Assumption 3 requires the operator-norm and cross-term quantities used for the certificate to dominate these terms. Summing gives $|\delta_i| \leq \sigma_i$.

The implementation proxy is not substituted for $G_\delta^{(i-1)}$ in this proof unless Assumption 3 validates that substitution.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound in Eq. (S7) yields $|\Delta_\psi(x, v)| \leq \sigma_m(x) + \sum_{j=1}^{m-1} c_j\,\sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) = \sigma_{\mathrm{total}}(x)$ for every admissible deviation input v. For $\epsilon_\kappa = 1$, the applied margin is $\sigma_{\mathrm{total}}(x)$.

Step 4 (Forward invariance). For v^* satisfying $A(x)v^* \leq b(x) - \sigma_{\mathrm{total}}(x)$, the definition $\hat{\psi}_m^{0}(x, v) = b(x) - A(x)v$ gives $\hat{\psi}_m^{0}(x, v^*) \geq \sigma_{\mathrm{total}}(x)$. Hence $\psi_m^{\mathrm{true}}(x, v^*) = \hat{\psi}_m^{0}(x, v^*) + \Delta_\psi(x, v^*) \geq \sigma_{\mathrm{total}}(x) - |\Delta_\psi(x, v^*)| \geq 0$. Because $x(0) \in \mathcal{C}$, all lower-order recursive conditions in the main article's definition of $\mathcal{C}$ are satisfied initially. The HOCBF forward-invariance theorem cited there therefore renders $\mathcal{C}$ forward invariant. □

7.3 Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\mathrm{ctrl}}(x) = \beta \sum_j \left| \frac{\partial (L_g L_{\hat{f}}^{m-1} h)}{\partial x_j} \right| \sigma_{\mathrm{GP},j}(x) \cdot u_{\max} \tag{S8}$$

where $u_{\max} = \sup_{u \in \mathcal{U}} \| u \|$. This term is evaluated separately for each barrier row and included in that row's total margin; Theorem 1 does not assume a fixed ratio between σ_{ctrl} and σ_{total}.